

Surname

Name

University ID Number

Signature

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- Write the solutions in the blank areas (use the back of the page, if needed)

- The number of pages is 4. Additional papers will not be considered.

- Clarity, order and precision are strongly considered for the final evaluation.

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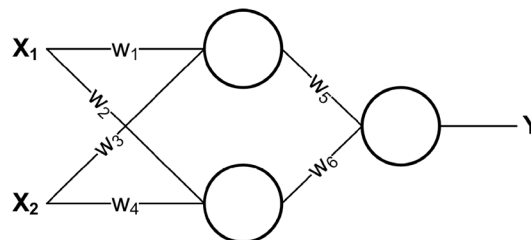
1. [Neural networks]

Consider a two-layer neural network to learn a function $f: X \rightarrow Y$, where $X = (X_1, X_2)$ consists of two attributes. The weights can be arbitrary. There are two possible choices for the function implemented by each unit in the network: L (linear function) and S (signed sigmoid function), so that, given the score $s = \sum_i w_i X_i$:

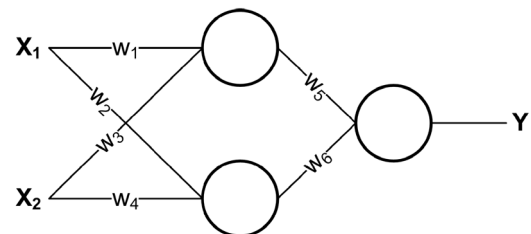
$$L(s) = cs,$$

$$S(s) = \text{sign}(\sigma(s) - 0.5) = \text{sign}\left(\frac{1}{1 + e^{-s}} - 0.5\right).$$

- a. Assign proper activation functions (S or L) to each unit in the following graph, so that the network simulates a linear regression:



- b. Assign proper activation functions (S or L) to each unit in the following graph, so that the network simulates a binary classifier:



- c. Show how a multi-layer perceptron can be used to represent the OR and the AND logical operators.
- d. Can the perceptron learning algorithm be used to train multi-layer perceptrons? Why?

Solutions

- a. Each unit must be linear (L).
- b. The inner layer must be linear (L), whereas the final layer needs to be sigmoidal (S).
- c. See the lecture notes (chapter 09).
- d. Only if there is only one perceptron in the final layer and the other nodes are linear. Moreover, PLA would terminate only if the data are linearly separable.

2. [Nonparametric classification]

The following data set shows classification data with two classes: plus (+) and minus (-).

-	-	+	+
	-		-
-	-	+	+

- Consider two nearest-neighbours (nn) classifiers based on standard Euclidean metrics: 1-nn and 3-nn. Describe their structures and show how each point would be classified in the two cases.
- Which one of the two has the largest Leave-One-Out Cross-Validation (LOO CV) error on this data set?
- Discuss the choice of k in k -nn classification.

Solutions

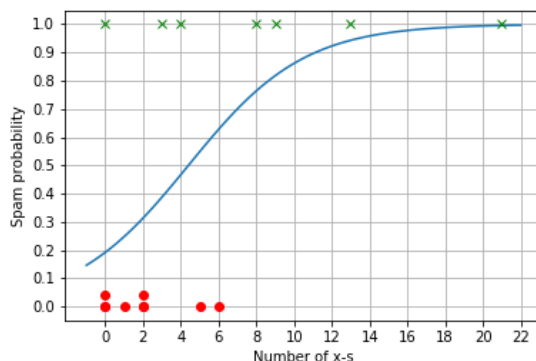
- See the lecture notes (chapter 06).
- The answer is 1-nn, since the 1-nn cross-validation error is 5/10 (all samples of the right cluster would be wrongly classified), whereas 3-nn cross-validation error is 1/10.
- See the lecture notes (chapter 06).

3. [Logistic regression]

You are building a spam filter. You started from the hypothesis that counting the occurrences of the letter 'x' in the e-mails will be a good indicator of spam or no-spam. You collect 7 spam messages and 7 no-spam messages and count the number of x-s in each. Here is what you find.

- Number of 'x'-s in each spam: [0, 3, 4, 8, 9, 13, 21]
- Number of 'x'-s in each no-spam: [0, 0, 1, 2, 2, 5, 6]

You train a logistic regression classifier on the data and plot the classifier against the data:



- Consider an e-mail with no 'x'-s. According to the model, what is roughly the probability of this message being a spam message and what is the probability of it not being a spam?
- How many x-s must an e-mail contain to guarantee it is a spam mail?
- How is a logistic regression model normally turned into a binary classifier? If you turn the model into a classifier in this way, what is the accuracy of the classifier on the training data?
- It is most important that no no-spams are classified as spams. How can this goal be described in terms of precision and recall? How can the logistic regression classifier be modified to try to achieve this goal?

Solutions

- We see from the graph that $P(1|x = 0) = 0.2$ and $P(0|x = 0) = 0.8$.
- You can never be 100% sure with a logistic regression model.
- This is normally done by choosing the class 1 if $P(1|x) > 0.5$. We see from the graph that this classifies 4 spams correctly and 3 spams incorrectly and 5 no-spams correctly and 2 incorrectly. Altogether 9 out of 14 are classified correctly, yielding an accuracy of 9/14.
- We could either say that the goal is to get a good precision for spam, or a good recall for no-spam. We achieve this by raising the threshold from 0.5. For the training data, a threshold of 0.7 would suffice. If we want to be prepared for more variation in the test data, we could set the threshold even higher.

4. [Principal Component Analysis]

An application of unsupervised learning is dimensionality reduction. Sometimes, however, it may be necessary to project data from a lower-dimensional space to a higher-dimensional space. Can PCA be used to generate higher dimensional representations of the dataset? Briefly explain why/why not and describe the main steps of the PCA algorithm.

Solutions

PCA cannot learn high-dimensional representations: intuitively, we cannot find more orthogonal dimensions than the ones provided; formally, the number of eigenvectors is limited by the dimensionality. See the lecture notes (chapter 12) for the description of PCA.